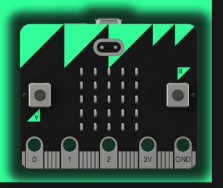
Introduction

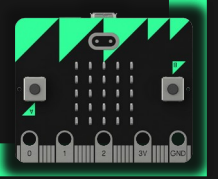
# The BBC Micro: Bit

Enniskillen Royal Grammar School



# Learning Objectives

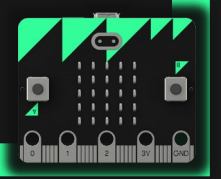
- Know what a BBC Micro: bit is
- Understand the different parts of the device
- Use the BBC Micro: bit website to create a simple program



https://makecode.microbit.org/

# What is the BBC Micro:Bit?

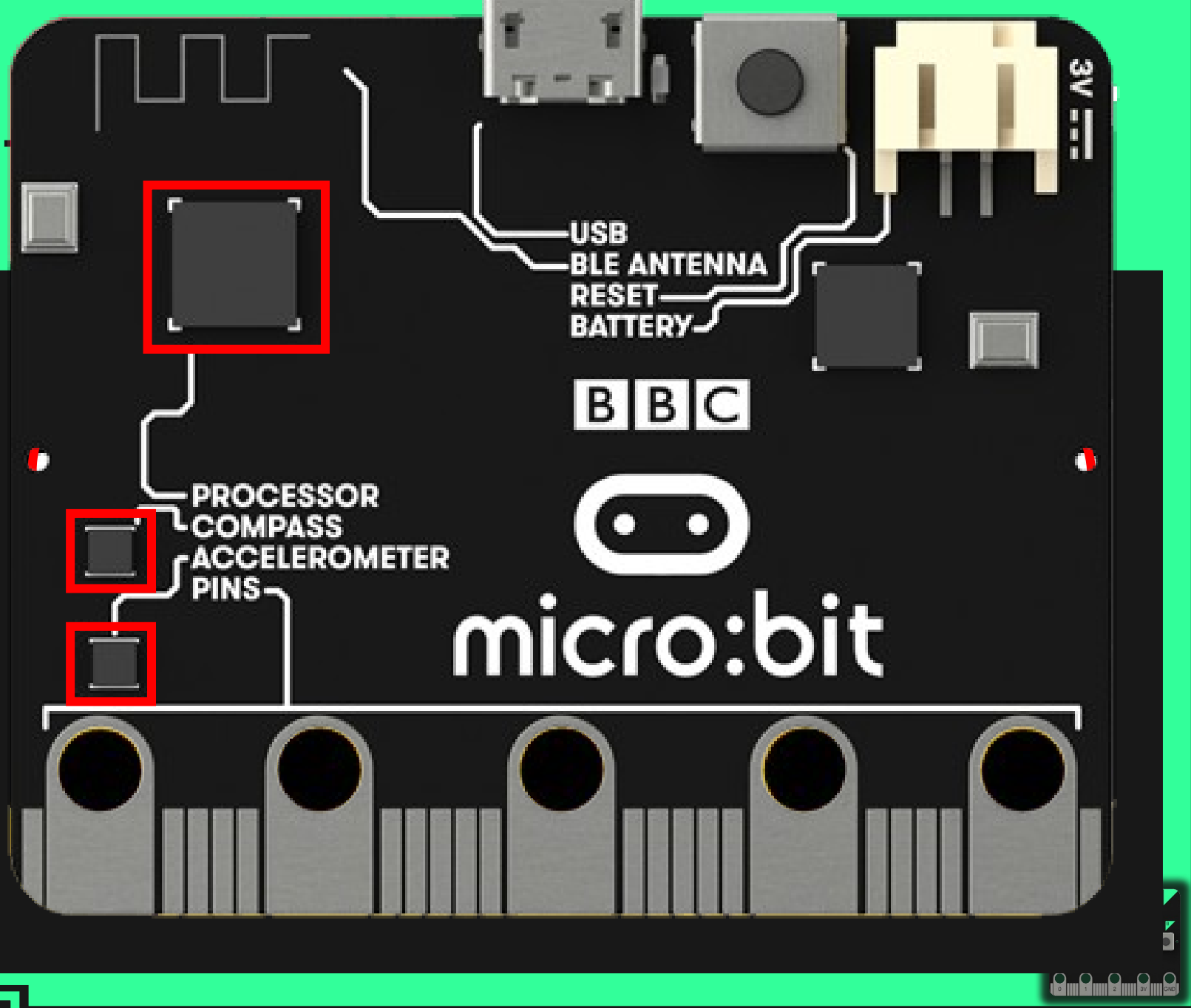
- Pocket Sized codable computer
- 25 red LEDs that can light up, flash messages, create games and invent digital stories
- Two programable buttons.
  - Can use it as a games controller
  - Can even be used to pause or skip songs on a playlist!
- Has a motion detector (accelerometer) that can detect movement
- Built in compass (magnetometer)
- Bluetooth smart technology – can connect to other Microbits or devices



https://makecode.microbit.org/

# The Componen

- Programmable Buttons
- 25 LEDs
- Compass
- Bluetooth
- Accelerometer



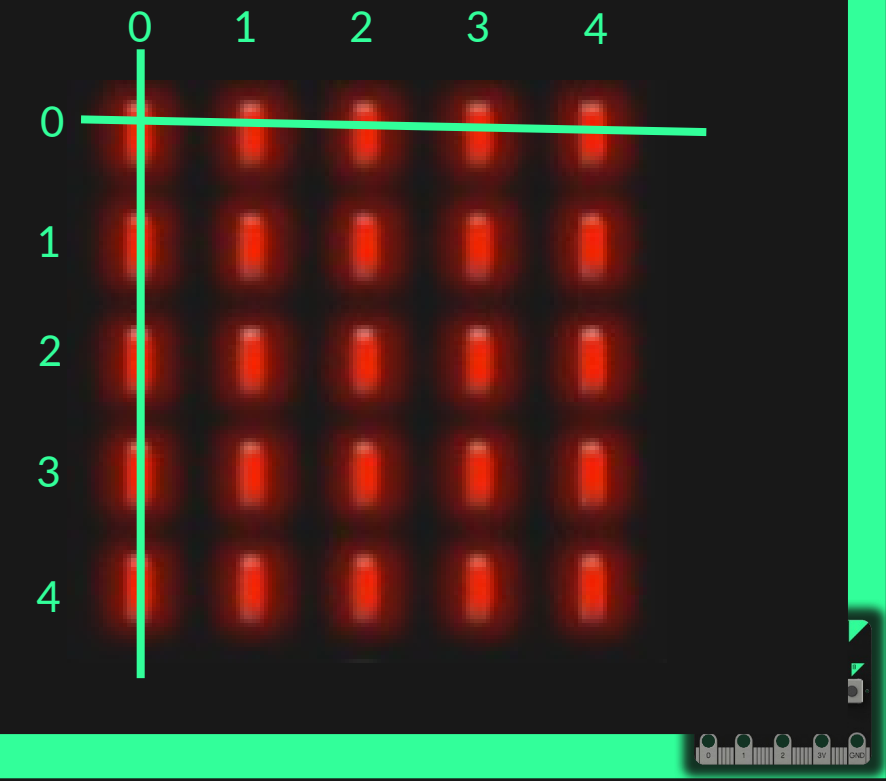

https://makecode.microbit.org/

# How does the LED light array work?

- The LEDs are arranged in a 5x5 grid
- Much like coordinates, these can be referenced to X and Y.
- Unlike coordinates, we reference them like so:

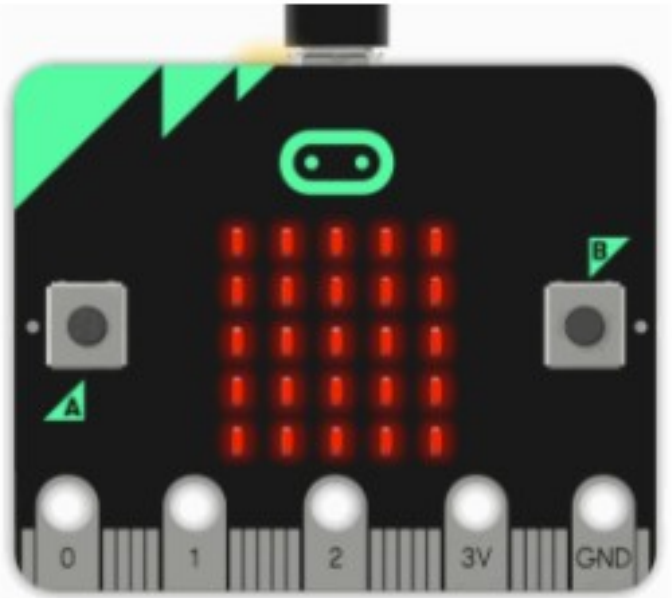
This LED would be referenced as

(3,1)



# Step 1: Connect It

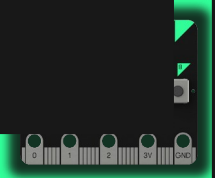
Connect the micro:bit to your computer via a micro USB cable. Macs, PCs, Chromebooks and Linux systems (including Raspberry Pi) are all supported. It comes with a fun application, give it a try!

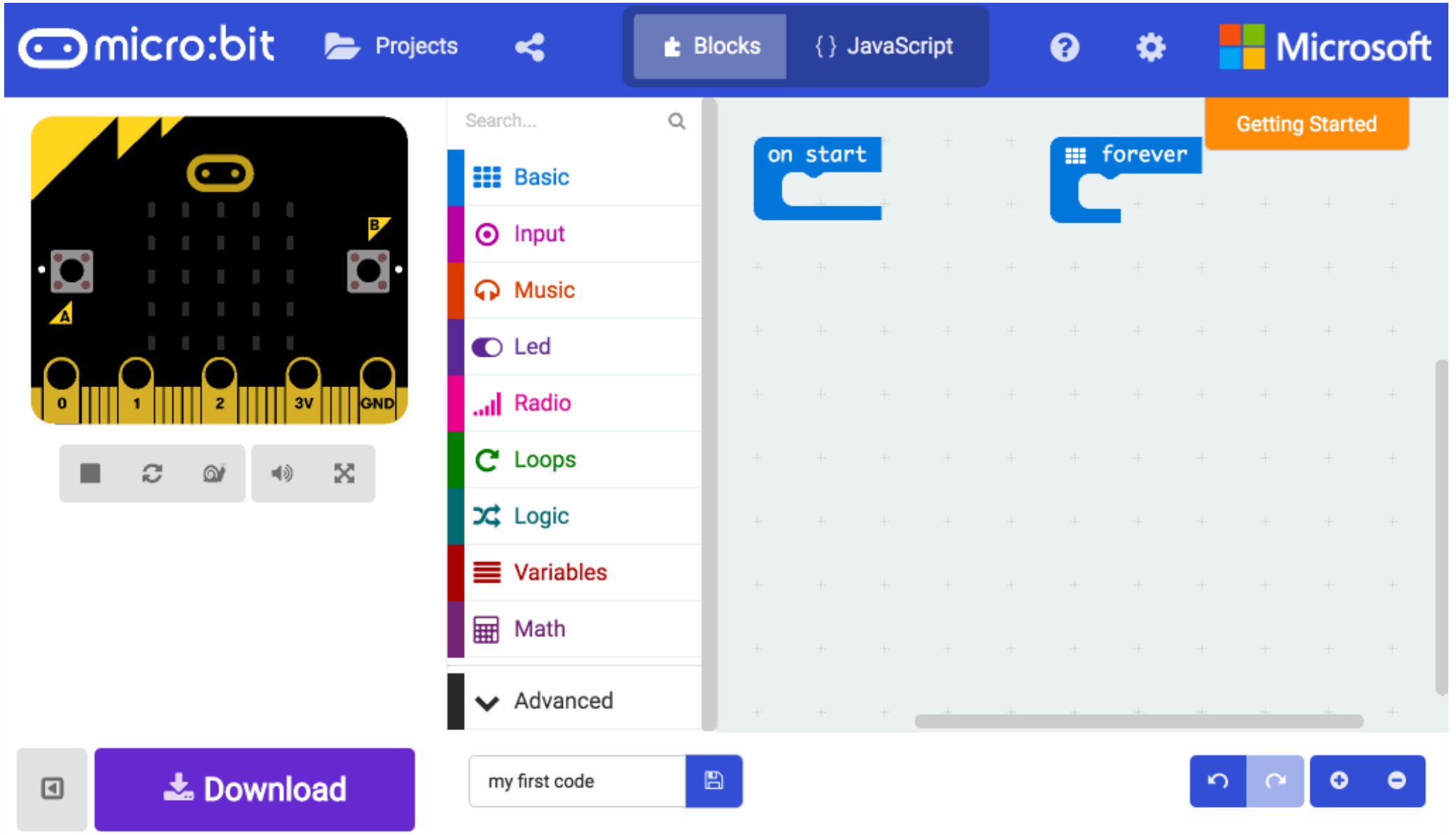



Your micro:bit will show up on your computer as a drive called 'MICROBIT'. Watch out though, it's not a normal USB disk!

# Website

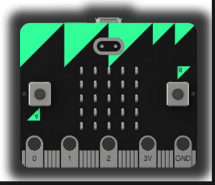
https://makecode.microbit.org/

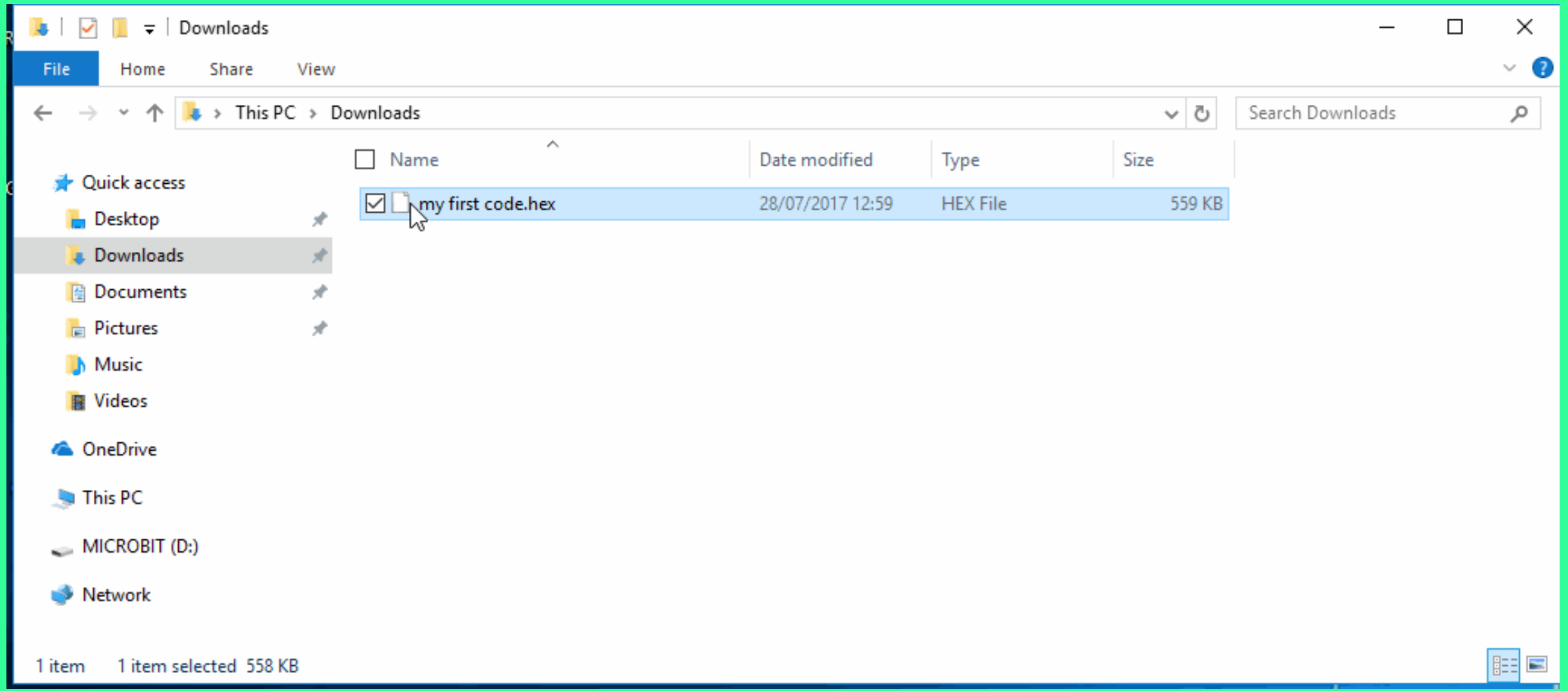



micro:bit
Projects
Blocks
{ } JavaScript
Microsoft
Search...
Basic
Input
Music
Led
Radio
Loops
Logic
Variables
Math
Advanced
on start
forever
Getting Started
0
1
2
3V
GND
A
B
Download
my first code

# Step 3: Download It

Click the Download button in the editor. This will download a 'hex' file, which is a compact format of your program that your micro:bit can read. Once the hex file has downloaded, copy it to your micro:bit just like copying a file to a USB drive. On Windows you can right click and choose "Send To→MICROBIT."

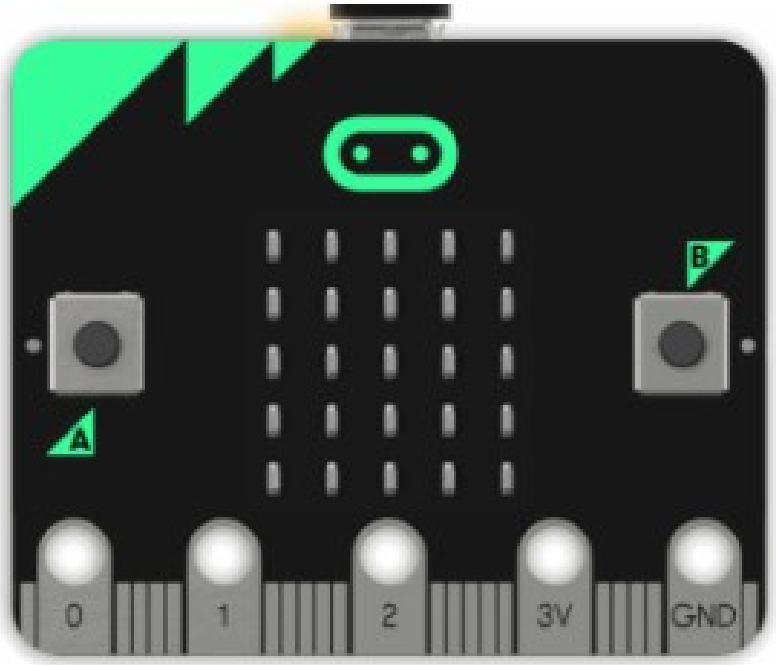





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# Step 4: Play It

The micro:bit will pause and the yellow LED on the back of the micro:bit will blink while your code is programmed. Once that's finished the code will run automatically!



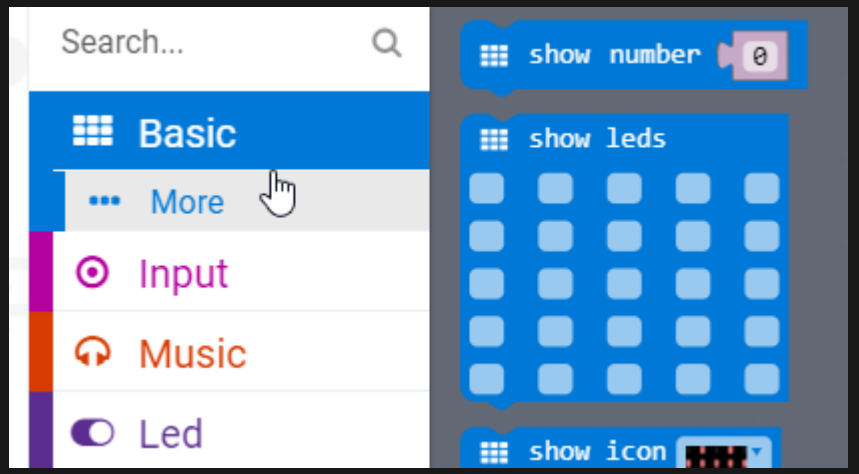

The MICROBIT drive will automatically eject and come back each time you program it, but your hex file will be gone. The micro:bit can only receive hex files and won't store anything else!

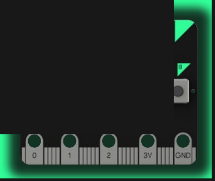
# Activity 1: Let's make an animation!

- Open up the Code Editor

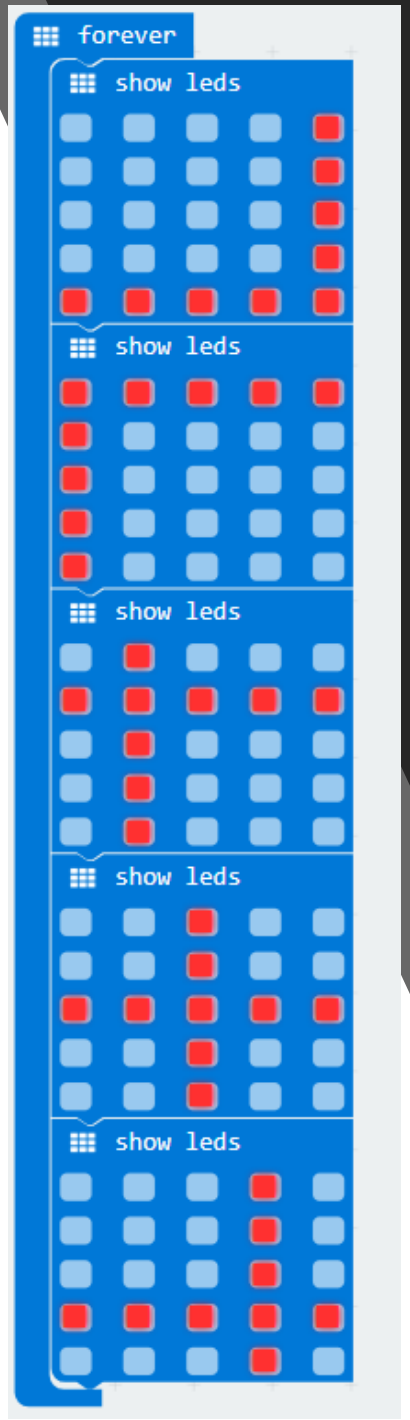



- Under **Basic**, use the **Show leds**



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# A Simple LED Program

- ← Create a program similar to this
- At the bottom of the screen you can give the program a name. Call this “MyFirstProgram”.
- Download and copy your hex file onto the microbit. The LEDs should loop this animation!
- Let’s Make another!

# What is a ‘string’?

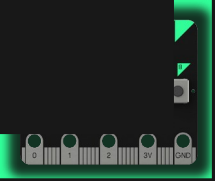
- A string is a piece of text.
- When we code, we have different types of **data** that a computer can understand:
  - Numbers
    - Integer
    - Decimal
  - Text
    - Character
    - String

A Whole number (positive or negative)

Numeric value with a decimal point

A single letter/symbol

When we have words/sentences etc



https://makecode.microbit.org/

# Let's make a program to say our name!

- Open up the website again
- Create a loop which uses



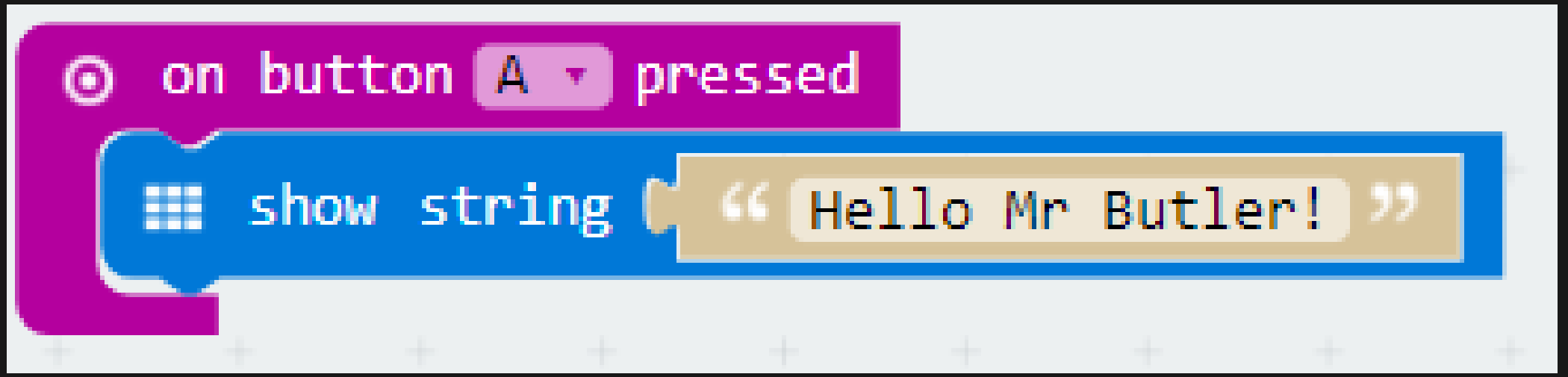

- Change the string value to “Hello Mr Butler”
- Download your code to the BBC Micro bit. Once it has copied across, does your message display on screen?

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# Programming Buttons

- With 2 programable buttons, we can use these as inputs for our Microcontroller.

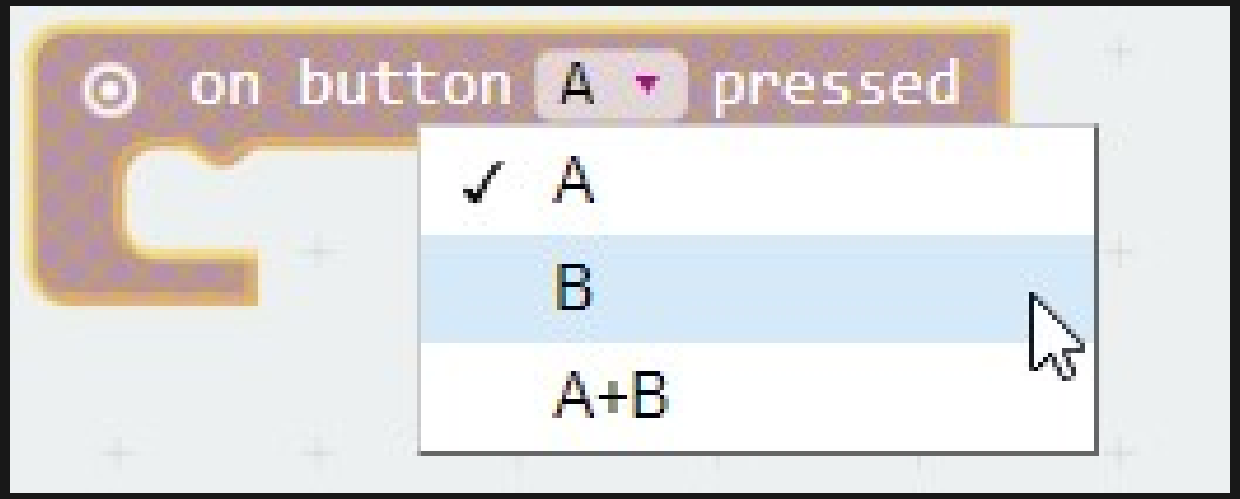
- Let’s make a short program:




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# Programming Buttons

- We can easily select this to be pressing the **B** button instead;

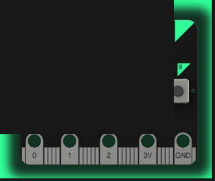



- Let’s see if you can complete Activity 2 now!

# Check Point!



- In your My Documents, you should have just Folders for each year.
  - Inside these, you should have Subject Folders with no orphan files!
- Within your ICT folder, there should now be a HMTL folder (instead of Mr Butler)
  - We have created a Micro Bit folder now too
    - buttonBashing – Hex File
    - MicroBit Portfolio – MS Word Document
    - smileyShake – Hex File

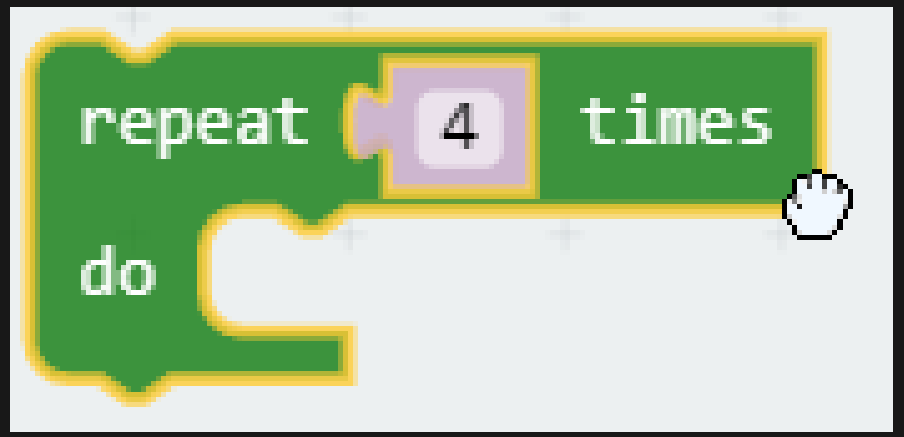


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# Looping

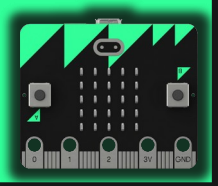
- Q: What does the word ‘Loop’ mean in programming?
- Ans: To repeat.

- For example, we could loop 2 or 3 times. Or 5. Or 100. Or even **forever**





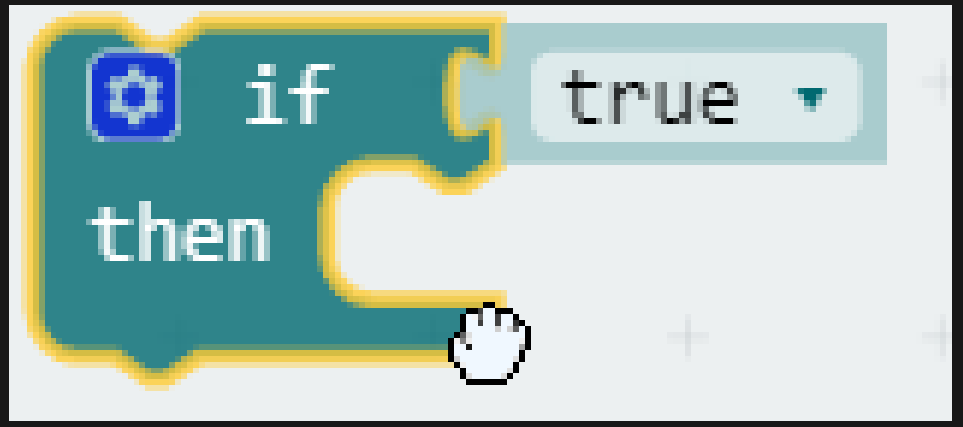

Save your Hex File into My Documents under **Looping**.

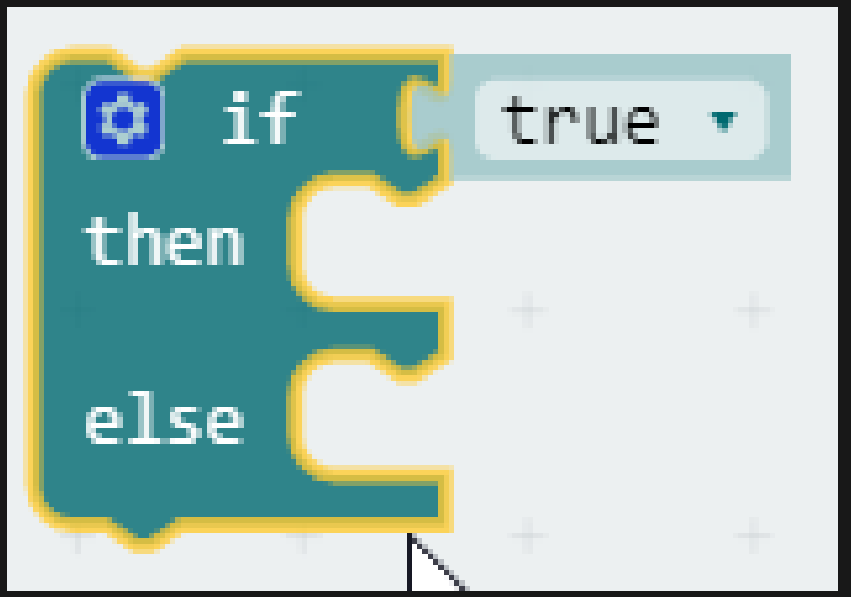


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# Using the IF function

- We have used the IF function in other areas of ICT. Can you name one?
- The IF statement uses the blue **Logic** blocks within the Micro: Bit programming environment.

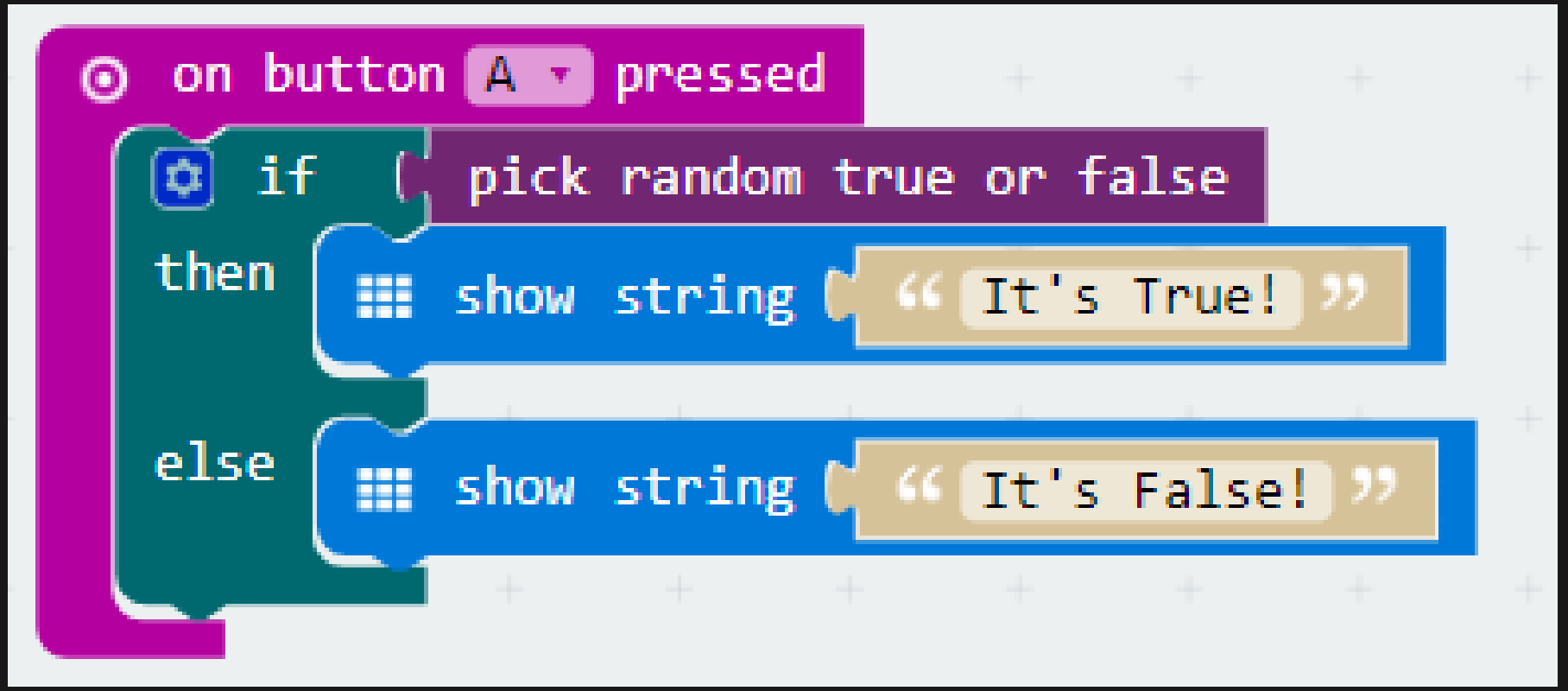



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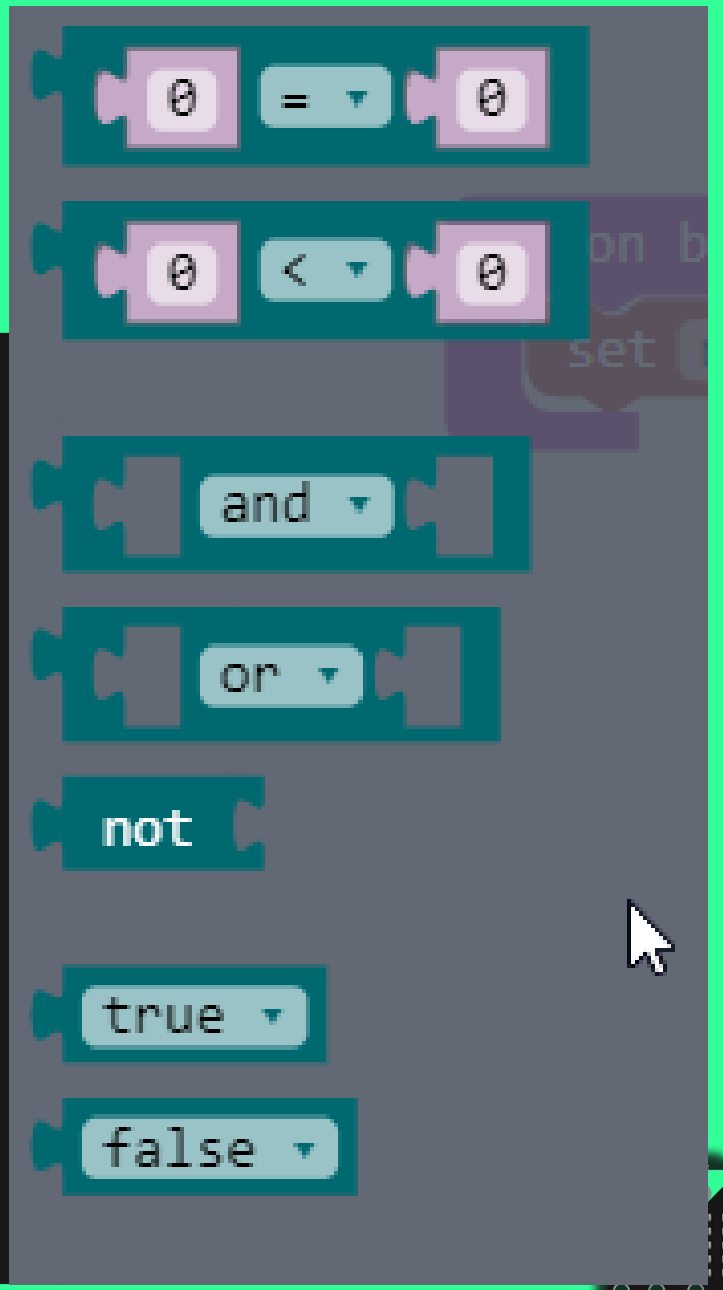
# The Lie Detector

- Create the script below. Save this as **theLieDetector** in My Documents



# Logic?

- There is a lot of logic blocks in the editor.
- Some of the blocks are on the right here.
- We can think **computationally** without even knowing about it
  - This means we can think like a computer
  - When computers run code, we have to tell them **exactly** what it needs to do as it cannot think for itself. We have to program the intelligence.
  - For example, when we make a cup of tea, what steps do we take?




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# >.< - The Difference

- You will notice we have a few different logical operators.
- This first one is checking if two values are equal



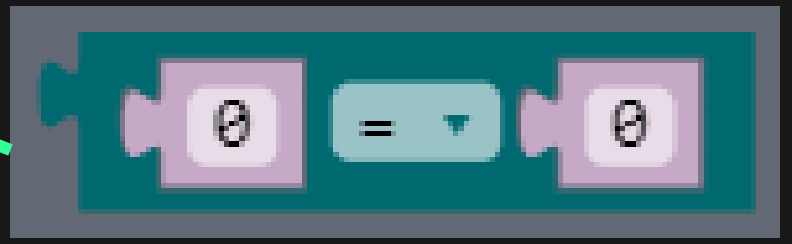

We also have ‘not equal’

Less Than

Less Than or Equal to

More Than

More Than or Equal to



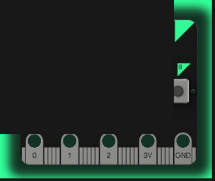



# Logical Operators

- We have had some previous experience with logic when we were working on our dice rolling activity.
- Logical operators have a true or false outcome.
- For example;

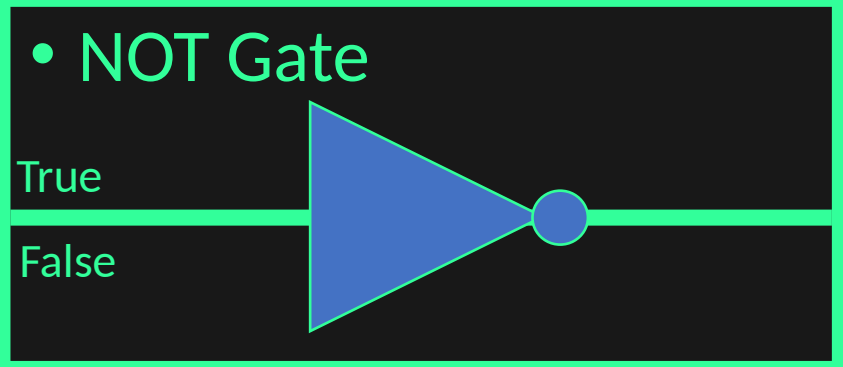
IF(5 > 2) ← This is true

IF(3 > 8) ← This is false

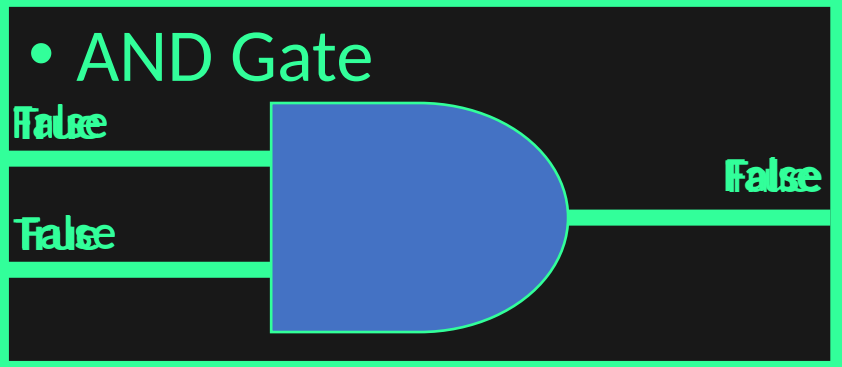


# Logic Gates

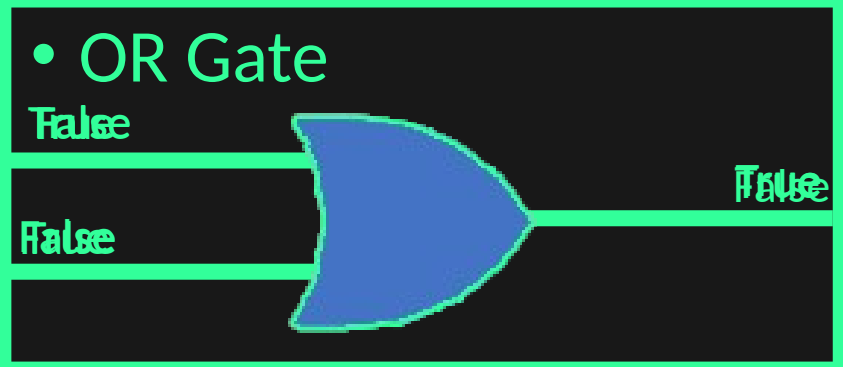
- NOT Gate



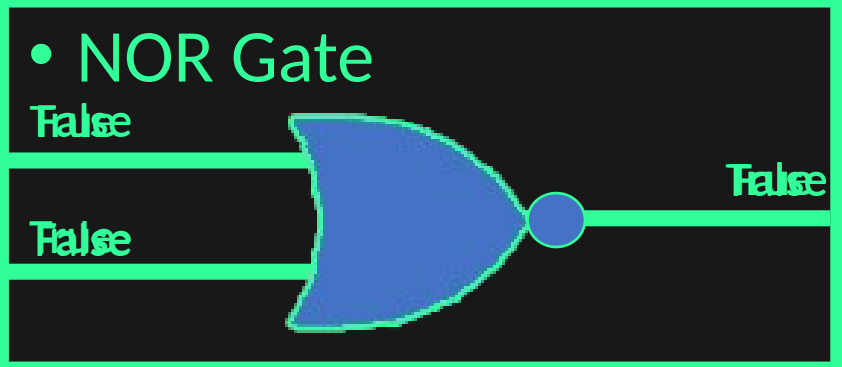

- AND Gate




- OR Gate




- NOR Gate




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